DEPARTMENT OF COMPUTER SCIENCE / INFORMATION TECHNOLOGY
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Web-Based Centralized Platform That Optimizes Food Waste Redistribution by Linking Restaurants with Registered Charitable Organizations

FINAL YEAR PROJECT (FYP) • BSCS SESSION 2024–2026 • SHAREMEAL PLATFORM

DESCRIPTION

Food wastage is one of the most critical socio-economic and environmental paradoxes in Pakistan. While millions suffer severe nutritional deprivation, banquet halls, restaurants, and catering services discard tons of wholesome surplus food daily due to the lack of an organized redistribution network. The **ShareMeal Platform** is an enterprise-grade web-based centralized system built on **ASP.NET Core 8.0, C# 12, Entity Framework Core, and Google Gemini AI**. The system establishes an instant digital bridge linking surplus food donors directly with certified charitable NGOs. By digitizing inventory, automating dispatch logistics, enforcing strict hygiene and expiry SOPs, and providing voice-enabled AI assistance, ShareMeal permanently eliminates food waste and feeds impoverished communities sustainably.

SCOPE OF THE SYSTEM

- **Nationwide Web Accessibility:** Responsive architecture operating seamlessly across desktop, tablet, and mobile browsers without requiring native app installs.
- **Multi-Tier Role Management:** Role-segregated secure portals for Food Donors (Restaurants), Claiming Charities (NGOs), and System Administrators.
- **Real-Time Food Inventory Matching:** Dynamic listings specifying portions (servings/kg), cooking timestamp, thermal state, and expiry countdown.
- **Atomic Concurrency Claim Engine:** Database-level transaction locking preventing duplicate claims or race conditions between competing charities.
- **Bilingual Multimodal Gemini AI (MealBot):** 24/7 intelligent voice-enabled assistant offering automated food safety SOPs, navigation, and community guidance.
- **Interactive Geolocation Mapping:** Integration with Leaflet.js and GPS routing for pinpointing donor addresses, route estimation, and distribution verification.

FUNCTIONAL REQUIREMENTS

- Secure user registration, authentication, and role assignment (ASP.NET Core Identity).
- Restaurant / Donor Module: Create, manage, edit, and track surplus food listings with countdowns.
- Charity / NGO Module: Live marketplace feed, geolocation filtering, and one-click instant reservation.
- Strict Verification Pipeline: Admin verification workflows verifying government NGO credentials and NTN.
- Status Lifecycle Engine: Automated transitions across *Available* → *Claimed* → *In-Transit* → *Delivered*.
- Food Safety & Hygiene SOPs: Pre-claim verification of food cooking time and storage temperature.
- MealBot AI Integration: Natural language processing in Roman Urdu, Urdu, and English with Web Speech API.
- System Analytics & Reporting: Comprehensive PDF and visual telemetry for meals saved and carbon offset.

METHODOLOGY (AGILE RAD MODEL)

The project adopted the **Rapid Application Development (RAD)** methodology under an Agile iterative lifecycle, emphasizing fast modular development, immediate stakeholder testing, and continuous deployment:

1. Requirements & Analysis
2. Architecture & Schema Design
3. Iterative Prototyping
4. QA & AI Integration
5. Cloud Deployment

COMPARATIVE ANALYSIS

Operational Feature	Manual Calling	WhatsApp Groups	ShareMeal Platform
Real-Time Matching	No	Partial	Yes (Instant)
Double-Claim Prevention	No	No (High Chaos)	Yes (Atomic Locking)
Verified NGO Authentication	No	No	Yes (Admin Vetted)
Food Safety & Expiry Tracking	No	No	Yes (Auto Countdown)
24/7 AI-Powered Assistance	No	No	Yes (Gemini AI MealBot)
Voice Accessibility (Bilingual)	No	No	Yes (Web Speech API)
Live GPS Route Mapping	No	Manual Links	Yes (Leaflet.js)
Audit Trail & Impact Reports	No	No	Yes (Automated PDF/Log)

RELATIONAL SCHEMA DIAGRAM

Organizations [Table]
- Id : int [PK]
- Name : nvarchar(150)
- Type : int (Donor/NGO)
- Status : int (Pending/OK)
- Address : nvarchar(250)
- City : nvarchar(80)
- Lat / Lng : float
- Phone / Email : nvarchar

Donations [Table]
- Id : int [PK]
- FoodItem : nvarchar(120)
- Quantity : int (portions)
- Status : int (0–4)
- ExpiryDate : datetime
- PreparationDate : datetime
- DonorId : int [FK]
- RecipientId : int [FK]

AspNetUsers [Table]
- Id : nvarchar(450) [PK]
- UserName : nvarchar(256)
- Email : nvarchar(256)
- PasswordHash : nvarchar(MAX)
- OrgId : int [FK]
- Role : nvarchar(50)

ContactMessages [Table]
- Id : int [PK]
- Name : nvarchar(100)
- Email : nvarchar(100)
- Subject : nvarchar(150)
- Message : nvarchar(MAX)
- CreatedAt : datetime

Figure: Normalized Relational Entity Relationship Diagram (1:M Referential Integrity)

TOOLS & TECHNOLOGIES

Hardware Tools

- **Processor:** Intel Core i7 / i5 Multi-Core
- **Memory:** 16GB DDR4 High-Speed RAM
- **Storage:** 512GB NVMe SSD
- **Display:** 1080p FHD / 4K Workstation
- **Testbed:** iOS Safari & Android Chrome

Software Tools

- **Backend:** ASP.NET Core 8.0 & C# 12
- **ORM & DB:** Entity Framework Core 8
- **Frontend:** Modern HTML5, TailwindCSS
- **AI Engine:** Google Gemini Flash API
- **Maps & Audio:** Leaflet & Web Speech
- **Cloud / DevOps:** Docker & Railway PaaS

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